

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024 (Course: B.Tech.)

Branch : Electronics and Telecomm. Engg./ Electronics and Comm. Engg./ Electronics Engg.

Subject Code & Name: BTHM705 Engineering Economics and Financial Mathematics Semester : VII

Max Marks: 60

Date: 21/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Firms ____ goods and services in the economy.	CO1	1
a) demand b) supply c) restrict d) consume		
2 The law of demand states that as the price of a good increases, the quantity demanded ____.	CO1	1
a) increases b) decreases c) remains constant d) fluctuates		
3 The equilibrium price is determined at the point where supply and demand ____.	CO1	1
a) diverge b) intersect c) decrease d) stabilize		
4 Sunk costs are expenses that cannot be ____.	CO1	1
a) calculated b) recovered c) reduced d) controlled		
5 The ratio of function to cost in value engineering is called:	CO2	1
a) Performance b) Value c) Efficiency d) Reliability		
6 A company decides to buy instead of making when it lacks ____.	CO2	1
a) Space b) Equipment c) Demand d) Experience		
7 The sinking fund factor helps in ____.	CO2	1
a) Saving b) Spending c) Lending d) Borrowing		
8 A revenue-dominated cash flow focuses on maximizing ____.	CO2	1
a) Savings b) Costs c) Returns d) Taxes		
9 A project is accepted if its Present Worth is ____.	CO2	1
a) Zero b) Negative c) Positive d) Constant		
10 Which maintenance type is performed at regular intervals to prevent failure?	CO2	1
a) Corrective b) Predictive c) Preventive d) Breakdown		
11 Which maintenance approach allows assets to run until they fail completely?	CO3	1
a) Preventive b) Breakdown c) Condition-based d) Predictive		
12 Which method applies an equal amount of depreciation each year?	CO3	1
a) Declining b) Straight c) Sinking d) Sum		

Q. 2 Solve the following.		12
A) Explain the circular flow model of an economy with suitable examples.	CO1	6
B) Describe the law of demand and supply with a suitable example.	CO1	6
Q. 3 Solve the following.		12
A) Define the make-or-buy decision. Describe criteria for make and buy.	CO2	6
B) Ramesh wants to save money for his daughter's college education, which will cost Rs. 5,00,000 after 10 years. He has access to a fixed deposit scheme offering 6% annual interest. How much should Ramesh deposit today to ensure he has Rs. 5,00,000 after 10 years?	CO2	6
Q. 4 Solve Any Two of the following.		12
A) Define cash flow. Describe types of cash flow with examples of each.	CO2	6
B) Describe the concept of the Present Worth (PW) Method with example.	CO2	6
C) A renewable energy project has:	CO3	6
• Initial investment: Rs 5 crore		
• Annual income: Rs 1.5 crore for 5 years		
• Salvage value at the end: Rs 1 crore		
• Discount rate: 10%		
Calculate its present worth.		
Q. 5 Solve Any Two of the following.		12
A) Discuss different types of maintenance with examples.	CO3	6
B) Define economic life of an asset. What factors affect its determination?	CO3	6
C) What do you mean by capital recovery? Describe the concept of Challenger vs. Defender.	CO3	6
Q. 6 Solve Any Two of the following.		12
A) What is depreciation? Explain its significance in financial planning and asset management.	CO3	6
B) Describe the Straight-Line Method (SLM) of depreciation with an example.	CO3	6
C) A textile factory purchases a machine for Rs. 10,00,000. The depreciation rate is 20% per year. Calculate the book value of the machine at the end of 3 years using the Declining Balance Method.	CO4	6

*** End ***